

Goutham Raju Kosuru Srinivasa

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SKILLS

Languages: C#, SQL, Python, TypeScript, JavaScript, Java, HTML, CSS

Technologies and Frameworks: .NET Core, ASP.NET Core Web API, FastAPI, NestJS, Angular, React.js, Next.js

Database and Storage: PostgreSQL, SQL Server, MongoDB, Relational Database Design, Query Optimization

Cloud and DevOps: AWS (Lambda, API Gateway, DynamoDB, S3, SQS, Bedrock, SAM, CloudFormation, EC2), Azure (App Services, Azure DevOps, Application Insights)

AI / LLM Tools: LangChain, AWS Bedrock, Claude Code, GitHub Copilot, Cursor

Developer Tools: VS Code, Git, GitHub Actions, Postman, Docker

Development Practices: Agile/Scrum, SDLC, CI/CD, Automated Testing, Incident Troubleshooting

EXPERIENCE

Software Engineer

Feb. 2026 – Jul. 2026

Community Dreams Foundation

Remote

- Redesigned backend and data layer of a compliance-critical platform with a PostgreSQL schema enforcing state transitions and row-level security.
- Implemented RBAC with AES-256 encrypted storage and immutable audit trails, enforcing access at the database level.
- Built a 26-test automated suite validating data integrity and state transitions in a CI/CD pipeline via Docker and GitHub Actions.
- Built structured logging and observability dashboards to proactively detect data-integrity and workflow issues; onboarded 20+ beta users across three distinct roles.

Software Engineer

Feb. 2022 – Jan. 2024

Neudesic, An IBM Company

Bangalore, India

- Developed and maintained three connected customer, employee, and admin applications using C#/.NET Core, Angular, TypeScript, and SQL Server, building REST APIs and business workflows serving 2,000+ daily users.
- Diagnosed a high-impact SQL performance defect using SQL Server Profiler, identifying N+1 query patterns; rewrote the data access layer with batched JOINS and composite indexes, cutting API latency by 32%.
- Strengthened TypeScript API contracts across three codebases, reducing null/optional-field handling errors and cutting recurring production runtime errors by 40%.
- Used Azure Application Insights to diagnose and resolve production incidents end-to-end; improved CI/CD pipeline reliability by 25% through root cause analysis of recurring build failures, helping maintain 99.5% uptime.

Software Engineer Intern

Aug. 2021 – Oct. 2021

The Sparks Foundation

Remote

- Built a serverless, event-driven backend using AWS Lambda and API Gateway, designing a DynamoDB schema around application access patterns and GSIs to sustain 1,000+ daily transactions at sub-300ms p95 latency.
- Automated infrastructure provisioning using AWS SAM and CloudFormation, reducing manual environment setup from 45+ minutes to under 10 minutes and cutting provisioning time by 80%.

PROJECTS

Multi-Agent AI Document Processing Pipeline

- Architected a fault-tolerant, event-driven pipeline with three specialized agents (classify, extract, validate) orchestrated via SQS with dead letter queues and exponential backoff retry, ensuring failed messages never block downstream processing.
- Implemented idempotency checks so redelivered messages under SQS's at-least-once guarantee never process a document twice; used DynamoDB for agent state and audit persistence.

Secure Task Management System

- Built a production-ready full-stack application in an Nx monorepo with JWT authentication, database-enforced three-role RBAC, and a shared TypeScript DTO library eliminating contract drift between frontend and backend.
- Designed a multi-tenant PostgreSQL schema with row-level data isolation and comprehensive automated test coverage, deployed via a Docker-based CI/CD pipeline.

PUBLICATION

- Object Detection and Text Reading Device for the Visually Impaired, IRJMETS, 2022. Co-authored peer-reviewed research (Impact Factor: 6.752) on a real-time YOLOv3 and Tesseract OCR system deployed on Raspberry Pi 4, achieving 15+ FPS via TensorFlow Lite quantization.

EDUCATION

Arizona State University

Master of Science in Information Technology. With Distinction. GPA: 4.0/4.0.

Tempe, AZ

Dec. 2025